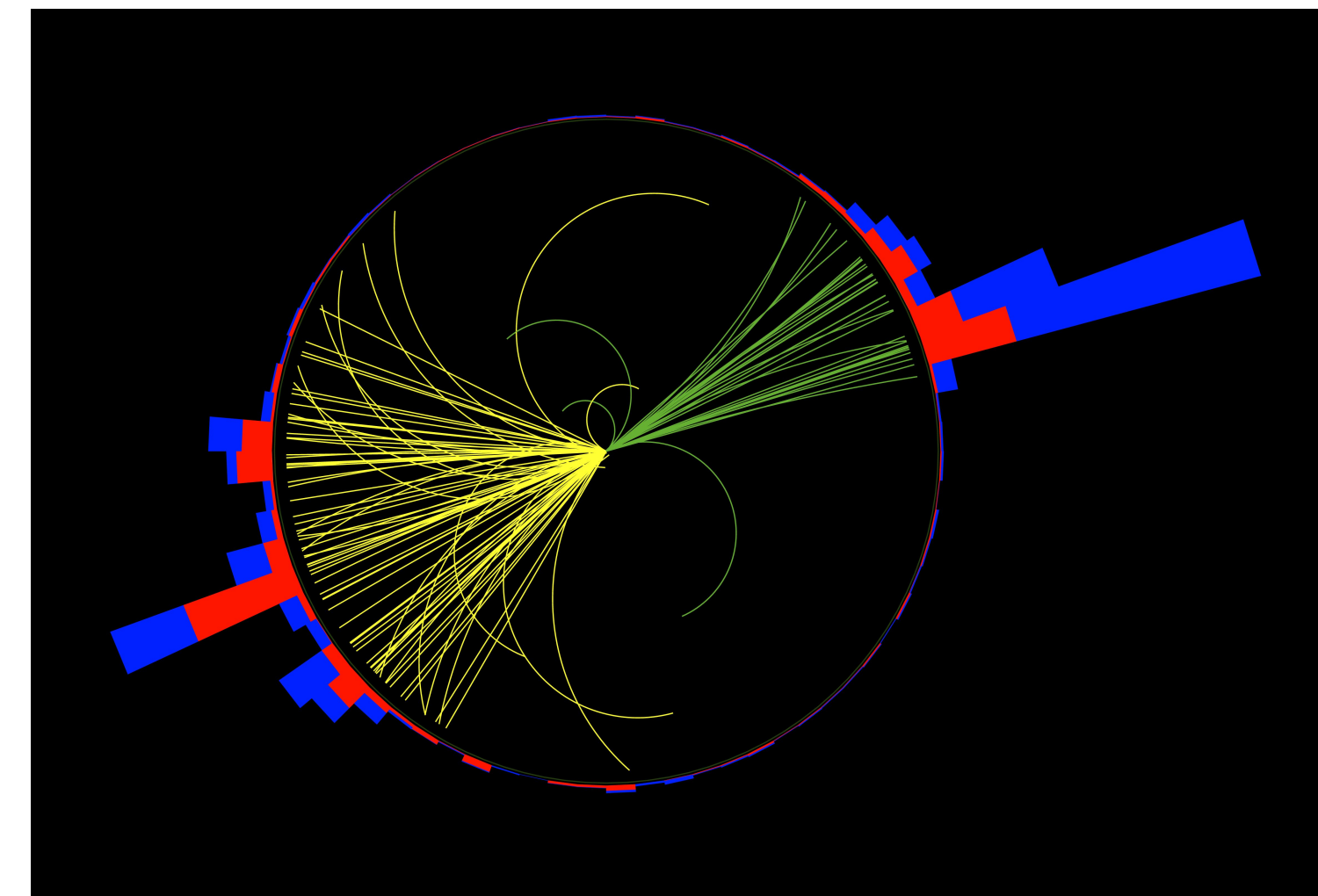
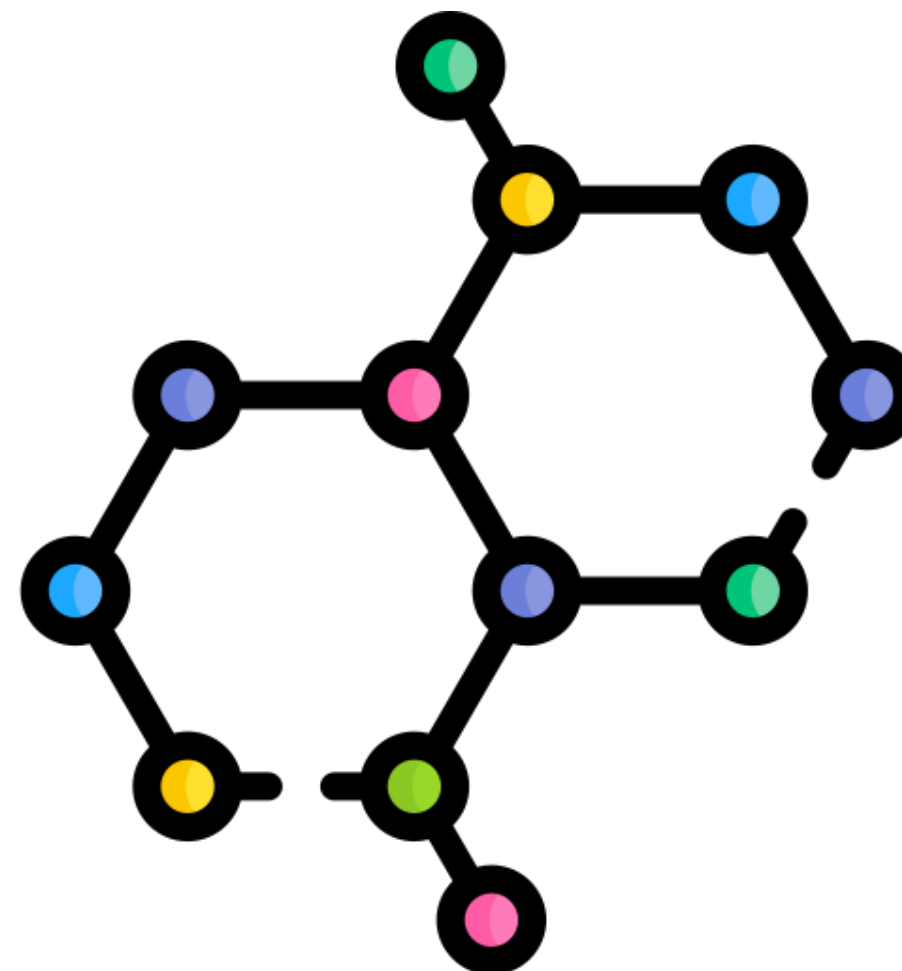
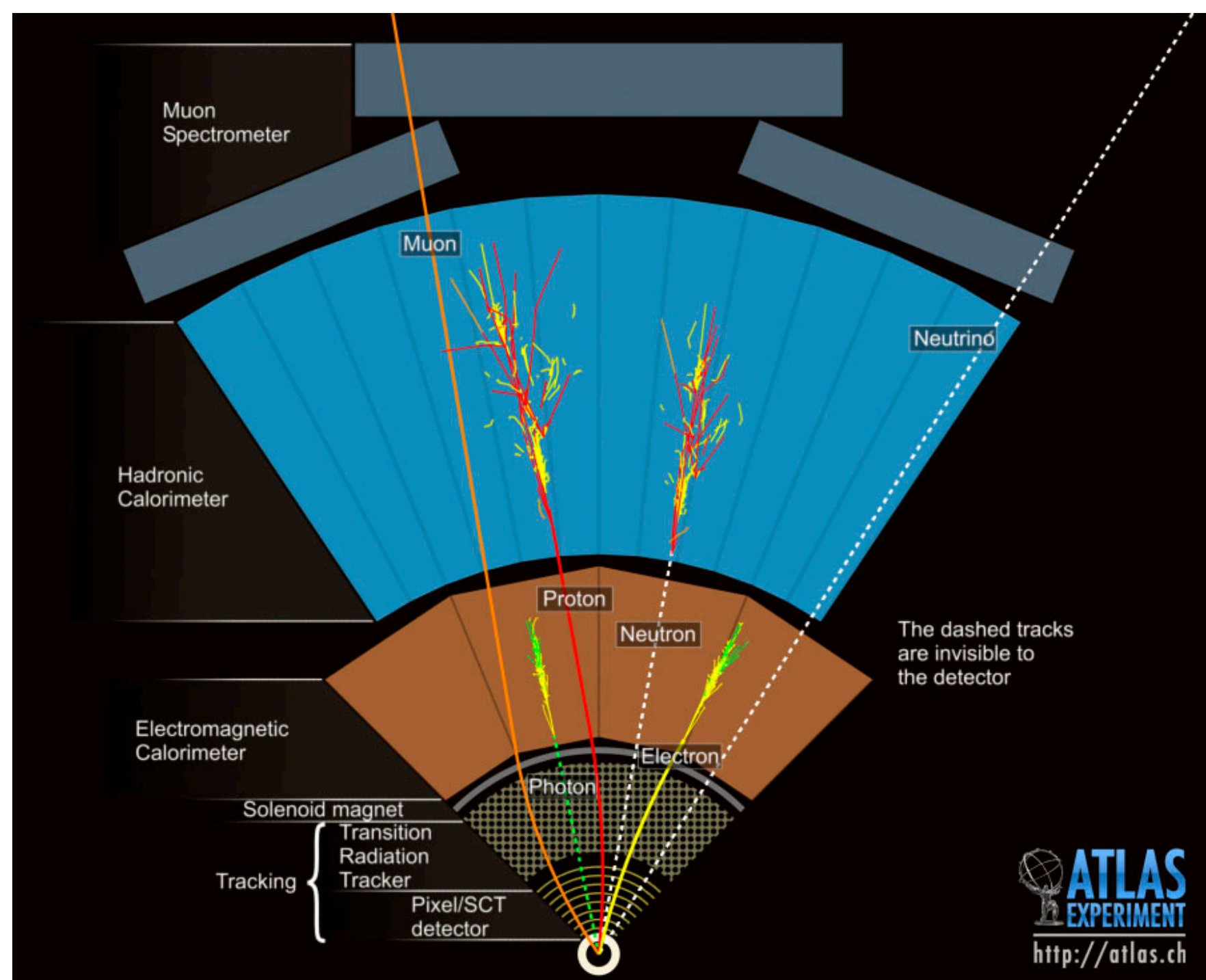




$$q\bar{q} \rightarrow e^+e^-\gamma$$



Attention is all you need .



0	1	1	1 _{×1}	0 _{×0}	0 _{×1}	0
0	0	1	1 _{×0}	1 _{×1}	0 _{×0}	0
0	0	0	1 _{×1}	1 _{×0}	1 _{×1}	0
0	0	0	1	1	0	0
0	0	1	1	0	0	0
0	1	1	0	0	0	0
1	1	0	0	0	0	0

I

*

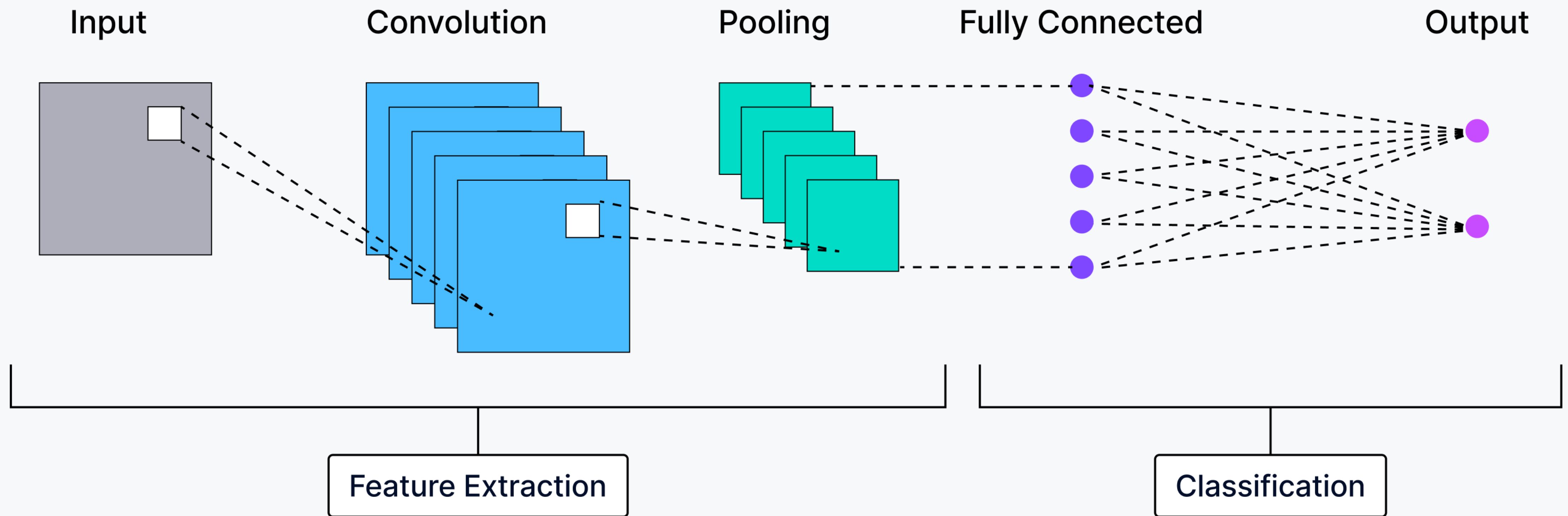
1	0	1
0	1	0
1	0	1

=

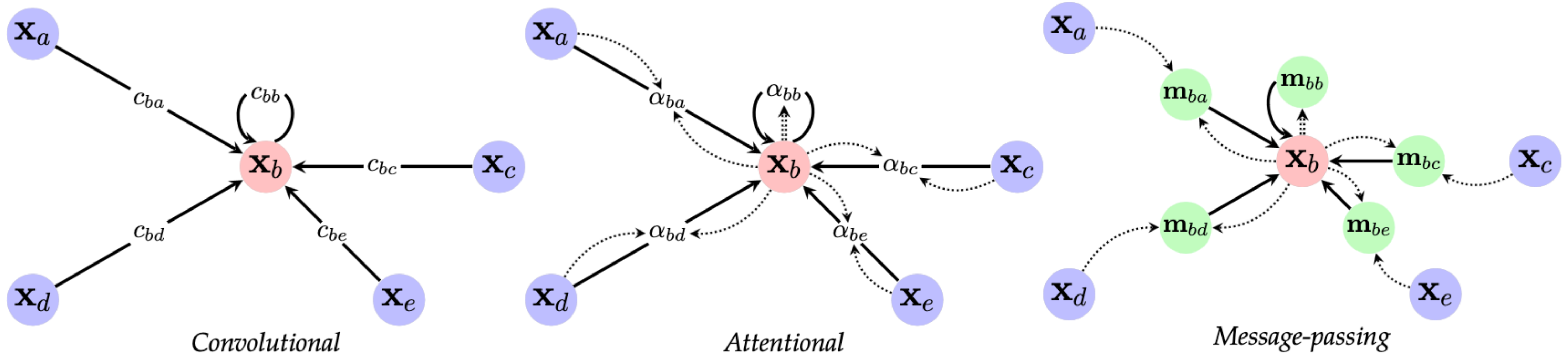
1	4	3	4	1
1	2	4	3	3
1	2	3	4	1
1	3	3	1	1
3	3	1	1	0

I * K

The Architecture of Convolutional Neural Networks



Types of graph networks



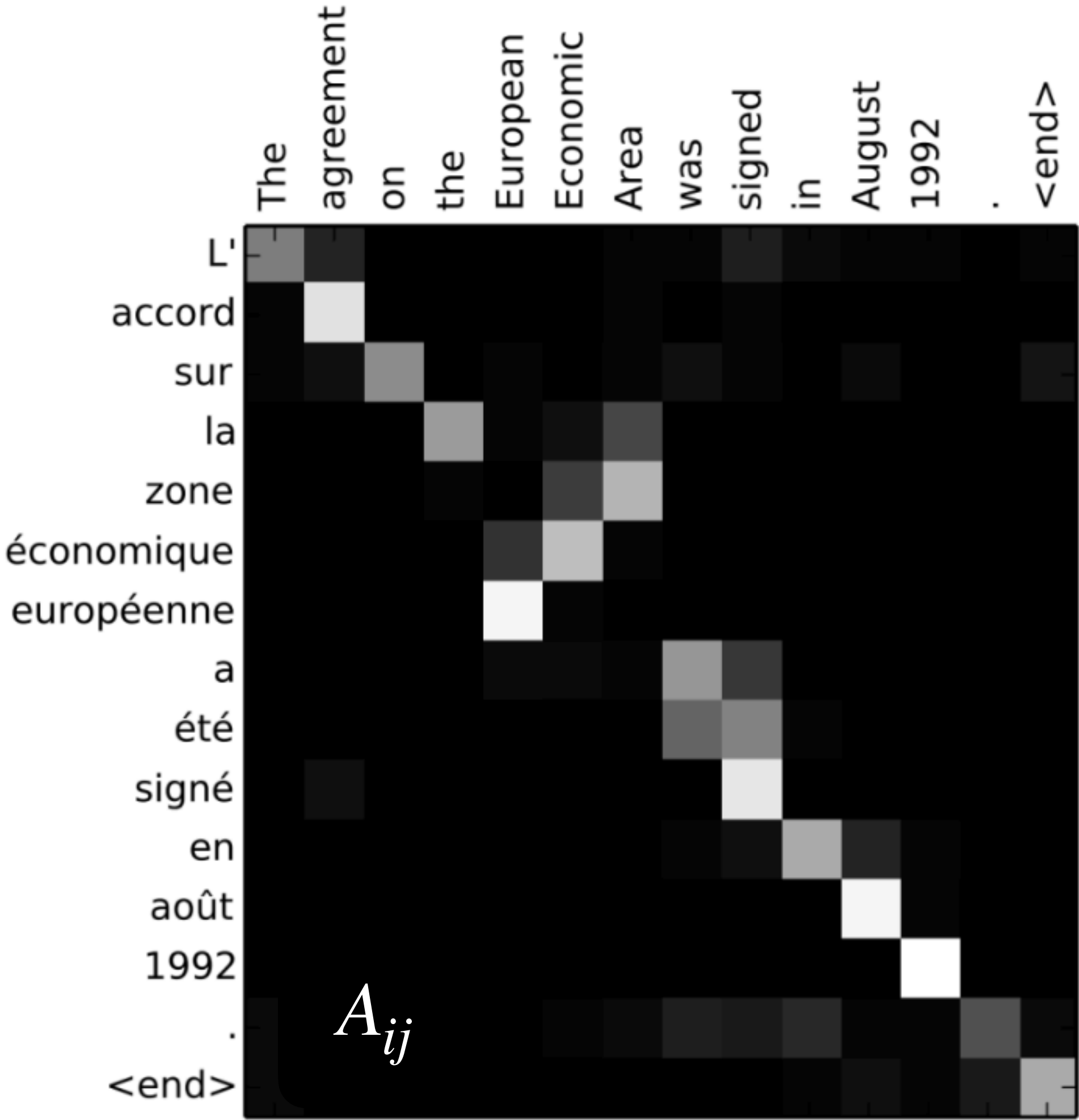
Self-attention

(columns=rows)

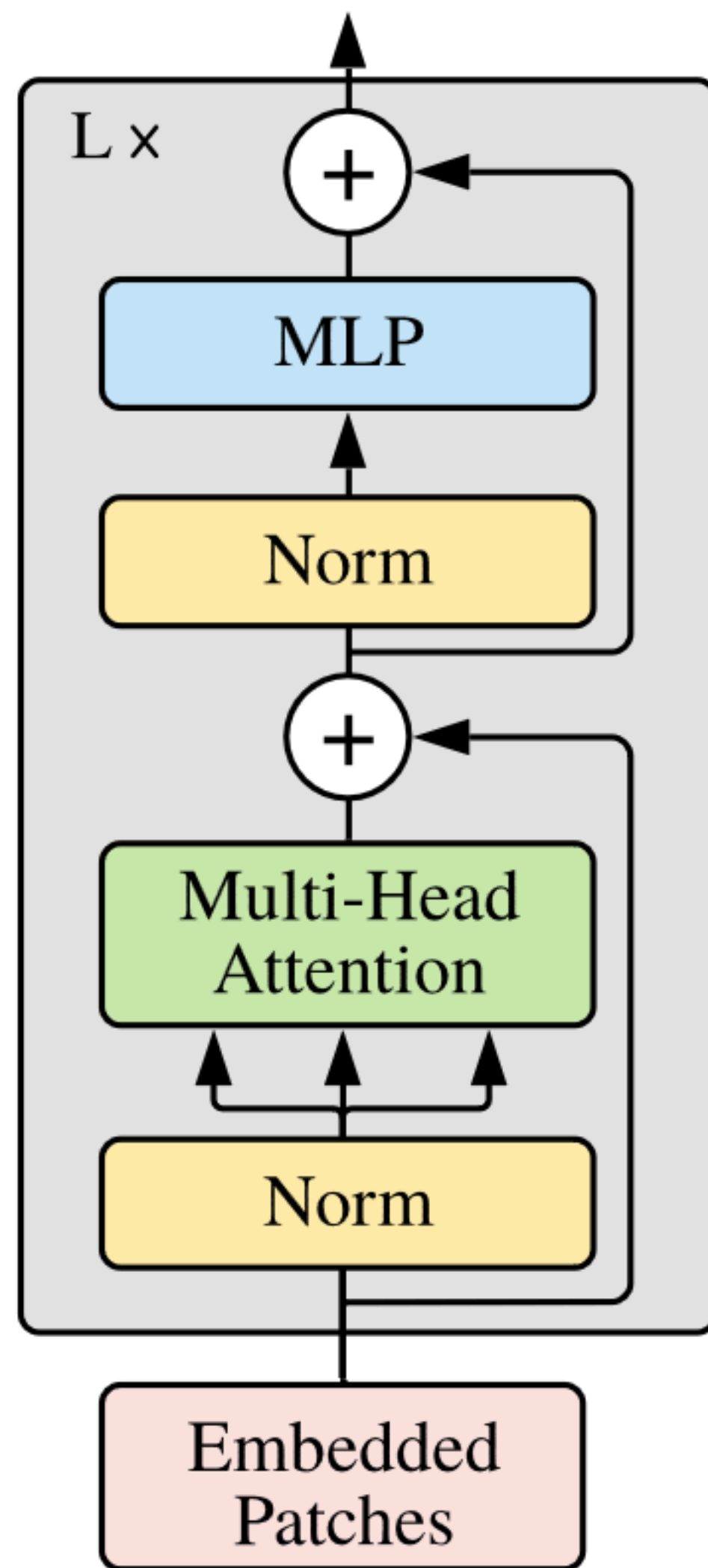
	Hello	I	love	you
Hello	0.8	0.1	0.05	0.05
I	0.1	0.6	0.2	0.1
love	0.05	0.2	0.65	0.1
you	0.2	0.1	0.1	0.6

Cross-attention

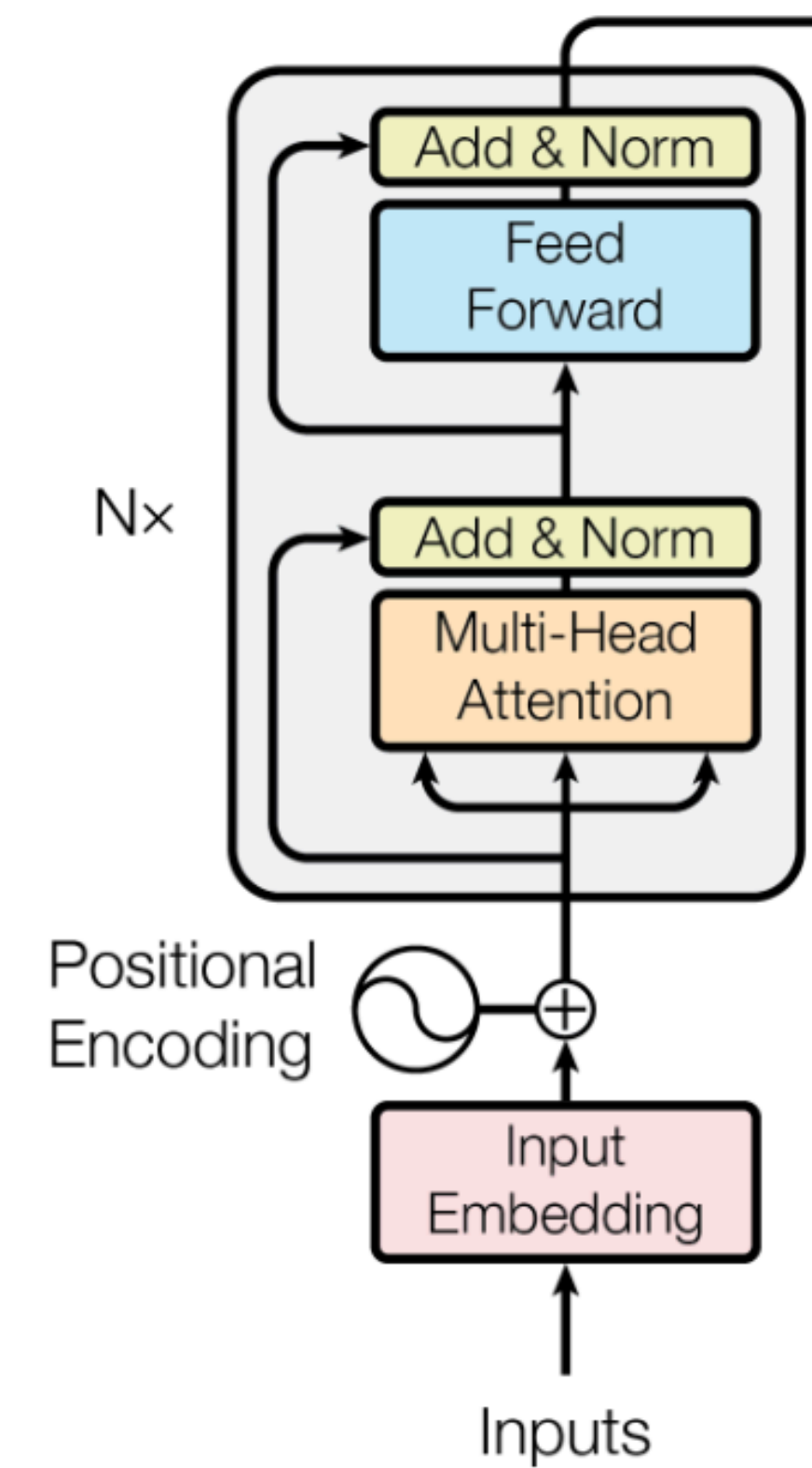
(columns≠rows)



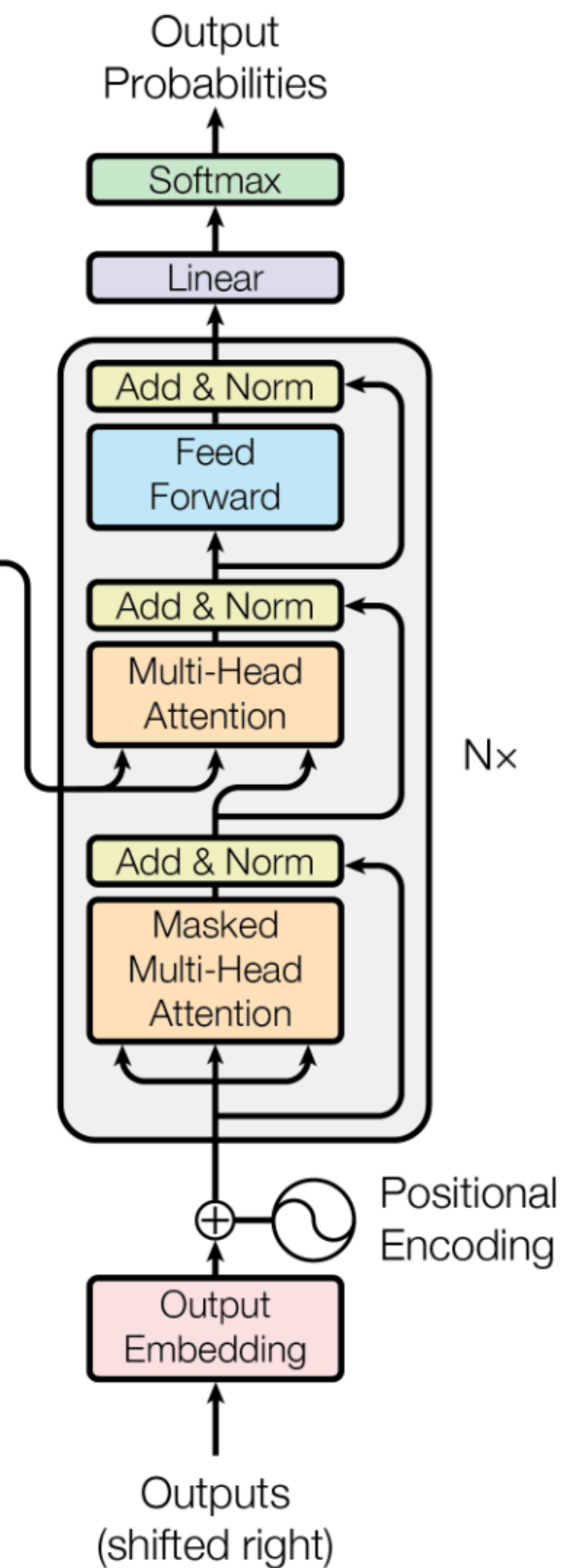
Transformer Encoder



Encoder



Decoder



Link to coding exercises:

github.com/spinjo/ml-lecture-ipp-2026